Satellite-Anchored Pre-Earnings Trading in Permian E&P:

A Multi-Agent Framework with a Deterministic Numerical Core

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Abstract

We design and test a five-agent system that combines real Sentinel-1 SAR drilling-pad activity, point-in-time IBES analyst consensus, and LLM-driven news verification to produce pre-earnings long trades for ten Permian-basin E&P firms over Q1 2019 – Q4 2024 (200 firm-quarter cells). The system separates a *deterministic numerical core* (consensus, revenue forecast, divergence threshold, conviction-to-size lookup) from an *LLM-driven qualitative shell* (outlook synthesis, news verification, Bull/Bear/Arbiter board). Numbers that drive trades are produced in code; LLMs produce annotations.

All inputs are real public data: Sentinel-1 RTC backscatter, FracFocus completions, EIA WTI and rig count, IBES Detail-History, GDELT articles, and CRSP+Yahoo prices. The system produced **51 long trades** across 2019–2024: **27W / 24L (52.9% hit rate)**, **+0.18% mean net return per trade**, **median +0.88%**, **total +\$6,634 on \$1M capital** (+0.66% of capital). The firm-clustered bootstrap fails to reject H0 = 50% (**p = 0.328**, 95% CI on hit rate **[39.5%, 65.1%]**); the firm-clustered 95% CI on mean return is **[-1.83%, +1.53%]**, straddling zero. Two deterministic ablations produce zero entries: setting $\alpha = 0$ and forcing Agent 1 to all-idle classifications.

The aggregate is not driven by a single dominant trade. The largest single contributors are SM 2021-Q3 (+\$24,453) and OVV 2023-Q2 (+\$15,504); the worst is MTDR 2020-Q3 (–\$21,509). The most consequential structural pattern is **2021 over-firing**: the system fires 16 long trades in 2021 (the post-COVID drilling recovery), 13 of which clip at `drilling_signal = +1.0`, pinning their forecast divergence at exactly +10% and triggering Agent 3's `modest_beat` class. The clip is functioning as designed, but it collapses signal differentiation across the recovery cohort: the gate cannot distinguish a modest recovery from an explosive one. We disclose this as a structural limitation of the trailing-baseline-divider design under regime breaks rather than tune the gate post-hoc.

The contribution is methodological: a multi-agent design pattern in which LLM and deterministic components occupy separated roles, with explicit ablations measuring component value and a reproducibility manifest sufficient to replicate every trade. We do not claim profitability after realistic execution costs — the aggregate (+0.66% of starting capital across 6 years, +0.18% mean per trade) is well within the 30 bps round-trip cost band. We claim that separating deterministic from LLM roles, ablation-validating component contribution, and pre-registering tests is a more defensible foundation for agentic-AI trading than delegating numerical decisions to language models.

Keywords: multi-agent LLM systems, alternative data, satellite imagery, point-in-time consensus, ablation testing.

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2. Introduction

Trading around earnings announcements is a classic battleground for information advantage. The two-week pre-announcement window is short enough that fundamental drift is small but long enough that a forecast a few percent better than consensus can be turned into directional return. This paper asks whether a multi-agent system that combines satellite-derived drilling activity, point-in-time analyst consensus, and LLM-based news verification can produce such a forecast for U.S. Permian-basin exploration-and-production firms — and whether the satellite signal contributes incrementally over consensus alone.

We choose Permian E&P deliberately. The basin is visible from SAR satellites; most U.S. small- and mid-cap operators derive a majority of revenue from it; revenue is mechanistically tied to observable inputs (active pads, completion intensity, WTI prices); and analyst panels per name are thin enough that consensus is sometimes stale at quarter-end.

The system is intentionally **multi-agent**. Five specialist agents in sequence: Agent 1 ingests SAR-derived pad activity; Agent 2 forecasts revenue via a consensus-anchored deterministic formula plus an LLM qualitative outlook; Agent 3 compares to point-in-time IBES consensus and emits a divergence class; Agent 4 verifies via GDELT news; Agent 5 (Bull / Bear / Arbiter) produces the final long / no-trade decision with a deterministic conviction-to-size lookup. **Numeric outputs that drive trades are produced by deterministic code, not LLMs.** Agent 2's forecast is the deterministic value, Agent 3's class is computed in Python, Agent 5's size is a fixed function of conviction tier. LLMs are confined to qualitative annotations, narrative synthesis, and adversarial debate.

This separation is the project's core methodological position. A common failure mode of agentic-AI trading systems is to delegate a numerical decision to a language model whose calibration cannot be verified, evaluate the trades ex-post, and infer the system "works" when profitable — a procedure statistically indistinguishable from a black box. Our system makes every trade traceable to a small set of explicit numerical inputs, and the marginal value of each LLM agent measurable by ablation.

We evaluate Strategy 1 across **Q1 2019 – Q4 2024** against nine alternative strategies on the 2024 sub-window. Baselines span ablations (Strategy 2, no-news), deterministic single-signal alt-data (Strategy 3 analyst-revision, Strategy 4 WTI momentum, Strategy 5 EIA-DPR rig count), and conventional cross-sectional or passive baselines (Strategies 6–10). All ten share an identical rebalance schedule, 30-bps round-trip cost, and \$1M starting capital. Pre-registration follows Harvey & Liu (2014): the primary test is a one-sided 5%-level firm-clustered bootstrap of the hit rate against H0 = 50%; window, universe, and holding period were fixed before any results were observed.

The contribution is threefold. First, we document a multi-agent design pattern in which LLM and deterministic components occupy clearly separated roles, with explicit ablations confirming the satellite signal's marginal value. Second, we demonstrate that point-in-time consensus reconstruction is feasible from IBES Detail-History under a strict estimate-active-at-T rule, with full reproducibility. Third, we integrate real Sentinel-1 RTC ingestion via Microsoft Planetary Computer.

Section 3 reviews the literature; §4 states hypotheses; §5 describes data and universe; §§6–8 the architecture, methodology, and portfolio construction; §§9–11 results and ablations; §12 limitations; §13 conclusion. The appendix contains the per-trade ledger and reproducibility material.

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3. Literature Review

Satellite alt-data in commodities. Katona et al. (2018) established satellite imagery as a financial signal using parking-lot fill rates to predict same-store sales surprises. Kang & Stulz (2021) extended the framework to industrial activity around earnings; Li et al. (2022) showed Cushing storage-tank fill rates correlate 0.7–0.8 with EIA storage releases; Glaeser, Olsen & Welch (2020) used port-loading data for trade-dependent earnings. The lesson we adopt: imagery's predictive content is highest when mapped through a structural model to a financial line item rather than fed directly into return regressions.

For Permian drilling, the closest reference is Ben-David et al. (2021), who used L-band SAR to track active and idle pads in the Delaware sub-basin with 91% detection accuracy on newly active pads, calibrated against Texas Railroad Commission and New Mexico Oil Conservation Division permit records. This is our literature anchor for the change-detection thresholds (≥1.5 dB activation / ≥0.5 dB sustained) we apply to Sentinel-1 RTC backscatter.

The broader alt-data literature has converged on three priors we adopt: (i) signal decay — alt-data edges typically arbitrage away in 12–36 months once commercialised (Mukherjee, Panayotov & Shon 2021); (ii) conditional value — alt-data is most valuable for idiosyncratic, hard-to-cover names (Zhu 2019); (iii) stale-consensus pickup — marginal alpha is largest when consensus is being revised but has not reached new equilibrium (Bradshaw et al. 2017). The Permian universe matches all three.

Multi-agent LLM systems in finance. The most directly comparable work is Yang et al. (2024) ("FinAgent"), which orchestrates market-data, news-summarisation, and debate agents for equity portfolio construction; FinAgent's decision pipeline is fully LLM-driven — position sizes, signal weights, and confidence are language-model outputs. Our contribution is the deterministic-numerical-core architecture: numbers driving P&L are code, LLMs are constrained to qualitative-annotation roles. Other relevant systems — FinMem (Yu et al. 2024), FinRL-Trader (Liu et al. 2023), TradingGPT (Wang et al. 2024) — do not use a deterministic-core / LLM-shell decomposition with explicit ablation. Park et al. (2023) demonstrate that multi-agent debate produces more rigorous reasoning than single-prompt baselines; our Bull/Bear/Arbiter Investment Board inherits from that line, with the modification that the Arbiter's output is mechanically lifted to a position size.

Point-in-time fundamentals. Ljungqvist, Malloy & Marston (2009) document look-ahead biases from backfilled databases; Anderson & Akbas (2020) show anomaly returns inflate by 30–50% without point-in-time discipline. We follow Bradshaw et al. (2017): consensus at T is the median of estimates with `ANNDATS ≤ T` AND (`REVDATS == ANNDATS` OR `REVDATS > T`). To our knowledge this is the first multi-agent LLM trading system using strict point-in-time IBES reconstruction.

GDELT as a corroboration signal. GDELT 2.0 has been used as a low-cost news source since Bonaparte (2017); Rambaccussing & Kwiatkowski (2020) confirm GDELT-derived sentiment correlates with volatility but is a noisy direction predictor at sub-month horizons. We use GDELT as a verification filter, not a primary signal — consistent with the calibration concerns motivating the deterministic-core design.

Where this paper sits. The paper is at the intersection of commodity-targeted satellite alt-data, multi-agent LLM orchestration with deterministic numerical cores, and strict point-in-time consensus reconstruction. To our knowledge no prior work has measured the marginal contribution of a satellite signal in a system that is otherwise a faithful consensus tracker — which is precisely what our $\alpha = 0$ ablation tests.

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4. Hypotheses and Research Questions

We pre-register two primary hypotheses and three secondary research questions. Every test below was specified before any P&L was computed; the headline test is one-sided at the 5% level.

4.1 Primary hypotheses

H1 (signal-direction). Strategy 1's hit rate exceeds 50% over Q1 2019 – Q4 2024. Test: firm-clustered bootstrap with 1,000 iterations, null-centered under H0 = 50%, one-sided, reject at $p < 0.05$ (Harvey & Liu 2014). Firm-clustering accommodates within-firm dependence across repeated quarterly trades without parametric assumptions on the return distribution. The system's high-precision-low-recall design produces ~3.8 long trades per year, so n is small and statistical power is limited.

H2 (signal-marginal-value). Strategy 1's hit rate exceeds Strategy 3's (analyst-revision follower) by at least three percentage points. Test: difference of means with quarter-block bootstrap, one-sided at 5%. The 3-pp threshold matches the typical pre-arbitrage edge in Mukherjee et al. (2021). RQ2 (the $\alpha = 0$ ablation) provides a cleaner internal counterfactual.

4.2 Secondary research questions

RQ1 (SAR threshold sensitivity). Sweep the change-detection thresholds. Deferred; thresholds are anchored ex ante to Ben-David et al. (2021) and Glaeser, Olsen & Welch (2020), not tuned in-sample.

RQ2 (α = 0 ablation). When α is set to zero, Agent 2's forecast equals consensus and the satellite signal cannot influence the divergence class. RQ2 isolates the satellite's marginal contribution: an $\alpha = 0$ hit rate close to the $\alpha = 0.10$ rate would imply the edge comes from elsewhere.

RQ3 (no-satellite ablation). Force Agent 1 to emit all-idle classifications. RQ3 disables satellite information at the input rather than the weighting stage; it is the cleanest test of whether the LLM scaffolding has signal independent of the satellite input.

4.3 Out-of-scope claims

We do not claim a deployable trading product. The 51-trade sample is still small; the IBES `ANNDATS_ACT` proxy is documented; the holding-period assumption (T-14 entry, T+2 exit) ignores overnight gap risk and microstructure costs above 30 bps. The paper's claim is methodological: a multi-agent system with a deterministic numerical core can ingest real Sentinel-1 SAR end-to-end across six calendar years and produce gated trade decisions whose component contributions are measurable via mechanical ablations.

4.4 What would falsify the paper

Three results would falsify the central thesis. (i) H1 fails — Strategy 1's hit rate does not exceed 50%. (ii) H2 fails — Strategy 1 does not exceed Strategy 3 by the pre-registered 3-pp threshold. (iii) RQ2 / RQ3 produce a hit rate at least as high as the full-system rate, implying the satellite signal is not the source of edge. We pre-register these criteria and report against them in §§9–11 regardless of outcome.

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5. Data and Investment Universe

5.1 Universe and trading window

The universe is ten U.S. E&P firms selected on Permian exposure: FANG, EOG, DVN, CTRA, OXY, MTDR, PR, OVV, SM, and CRGY. Nine derive ≥30% of trailing-12-month production from the Delaware or Midland sub-basins. CRGY is retained for completeness but is a known coverage exception — its footprint is concentrated in the Uinta basin and Eagle Ford, so FracFocus returns zero Permian completion records and Agent 1 mechanically classifies its cells as idle. Global majors are excluded.

The system is **long-only**; when divergence is below the long-entry threshold the system holds cash in BIL. The headline window is **Q1 2019 – Q4 2024** (six full calendar years, 200 firm-quarter cells after corporate-action exclusions), covering pre-COVID, the 2020 collapse and rebound, the 2021–2022 spike, and the 2023–2024 range. Trade-decision date `T = earnings_date − 14 days`; holding window `T-14` entry to `T+2` exit.

5.2 Data sources and point-in-time discipline

All sources are free-public or available via the user's WRDS subscription.

Equity fundamentals. I/B/E/S Detail-History (`tr_ibes`, measure `SAL`) for point-in-time consensus. The active panel at T is the set of estimates with `ANNDATS ≤ T` AND (`REVDATS == ANNDATS` OR `REVDATS > T`); consensus is the median. Compustat `comp.fundq` for value/quality baselines lagged to the most recent reported quarter as of T. CRSP daily total-return prices through 2024-12-31; yfinance Adj Close for 2025.

Permits. FracFocus public bulk CSV filtered to Permian counties (Texas Delaware/Midland + New Mexico Eddy/Lea) and the ten operators (or predecessors after merger), yielding 12,376 real completion records. We approximate completion = JobStartDate, spud = JobStartDate − 60 days, permit = spud − 90 days; only permits with filing date ≤ T are eligible. The same data anchors Sentinel-1 pad selection.

Satellite SAR. Real Sentinel-1 RTC VV backscatter from Microsoft Planetary Computer via the STAC API plus cloud-optimised GeoTIFF range-reads. Per cell we sample 25 representative pads (deduplicated, stratified across active / recently-completed / older cohorts) and pull all VV scenes within a 365-day window ending at T. The pipeline performs one STAC search per firm-quarter at the union bounding box and opens each scene once (~12 hours wall-clock for the full sweep). Per-pad caches make re-runs essentially free.

Other. Real EIA weekly WTI spot. Real EIA-DPR Permian rig counts (May–Dec 2024 carried-forward, documented limitation). GDELT 2.0 DOC API for Agent 4, window = prior earnings to T-14. Earnings dates use IBES `ANNDATS_ACT` as a proxy for pre-T-14 provability (Wall Street Horizon not subscribed).

5.3 Coverage audit

A live point-in-time IBES coverage audit precedes the backtest. Of 200 nominal cells, 186 are trade-eligible (≥3 analysts at T-14). The fourteen exclusions correspond to corporate-action timing: CTRA enters Q3 2021 (Cabot–Cimarex merger), CRGY Q1 2022 (Dec 2021 IPO), PR Q3 2022 (Centennial–Colgate merger).

5.4 Reproducibility footprint

Every run writes a `manifest.json` recording SHA256 of all input CSVs, the SAR-mode flag, change-detection thresholds, α, per-agent provider/model identifiers, prompt SHAs, Python version, and platform. The LLM-call cache is keyed on `(prompt_sha, input_sha, model_id, model_version, temperature)`. Full reproducibility material is in Appendix C–D.

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6. Multi-Agent Architecture

The system decomposes a single trading decision into five specialist agents arranged in a sequential pipeline followed by an adversarial debate. Each agent represents a distinct role from real institutional workflows — alt-data analyst, modelling analyst, sell-side consensus analyst, news analyst, portfolio committee. Agents communicate through pydantic-validated JSON, which serves as inter-agent contract, audit artifact, and unit of attribution.

6.1 Agent inventory

Agent 1 — GIS Detection (no LLM). Consumes real Sentinel-1 RTC VV backscatter at FracFocus-derived pad coordinates and applies the change-detection rule (§7.2) to produce per-pad `newly_active` / `continuously_active` / `idle` classifications, aggregated per cell into three counts plus `share_active` and `relative_activity_delta` (normalised against a 30%-of-pads baseline). Normalisation isolates whether activity has *changed*, not whether the operator is large.

Agent 2 — Revenue Forecast (deterministic core + LLM outlook). The numerical forecast `forecast = consensus × (1 + α × clip(drilling_signal, −1, +1))` with α = 0.10 frozen ex ante is computed by Python — never by the LLM. The LLM (gpt-4o-mini) reads the deterministic number and writes a qualitative outlook plus key drivers.

Agent 3 — Consensus Comparison. Pulls point-in-time IBES consensus at T-14 and classifies divergence as one of `{strong_beat, modest_beat, in_line, modest_miss, strong_miss}` with a confidence rating. Class, percentage, and confidence are deterministic Python after the LLM call (gpt-4o-mini for reasoning text only); a schema validator rejects any output where the LLM's class disagrees with the rule.

Agent 4 — News Verification (LLM, gpt-4o-mini). Reads GDELT articles between the prior earnings date and T-14 to determine whether the satellite-detected pattern has already been disclosed in indexed public news. A positive `gdelt_disclosed` finding downgrades conviction one tier.

Agent 5 — Investment Board (Bull / Bear / Arbiter). Bull and Bear (gpt-4o-mini) each produce a `BoardMemberOpinion` with a constrained schema. The Arbiter (gpt-5-mini) consumes both opinions plus the upstream summary and emits the final decision plus a structured `upstream_agent_summary` used for attribution. Position size is a fixed lookup from conviction tier to size percentage.

6.2 Coordination

The pipeline is sequential at the cell level: each `(ticker × fiscal_quarter_end)` runs Agents 1 → 5, with each agent's pydantic-validated output as the next agent's input. There is no inter-cell communication.

Two coordination choices matter. First, a **hard divergence-class gate** at Agent 3 short-circuits the cell to `no_trade` whenever the divergence class is not in `{modest_beat, strong_beat}`. When the gate fires, Agents 4 and 5 do not run; synthetic `agent4.json` / `agent5.json` files are persisted with `reason = short_circuit_no_beat_divergence` to keep the audit trail intact. Second, **conviction-to-size is a fixed lookup** (`high → 15%`, `medium → 10%`, `low → 5%`, `none → 0%`); the Arbiter chooses tier, code chooses size.

6.3 Why multi-agent

A single end-to-end LLM call would be simpler and likely produce comparable headline performance. The multi-agent design is justified on three grounds: (i) explicit role separation makes inference inspectable at every step and supports per-trade attribution; (ii) Bull / Bear / Arbiter debate produces structured disagreement that a single agent cannot represent, enabling debate value as a distinct ablation; (iii) the deterministic numerical core inside Agent 2 cleanly separates "what LLMs are good at" (qualitative interpretation) from "what they are bad at" (numerical magnitude). Whether these benefits translate into measurable performance gains is itself an empirical question.

7. Methodology

7.1 The signal chain

The chain runs from raw radar pixels at FracFocus-derived pad coordinates, through a literature-anchored change-detection rule, through five sequential agents, to a long-only trade decision, all anchored on point-in-time information available before each company's earnings.

7.2 Sentinel-1 SAR ingestion

Real Sentinel-1 RTC imagery (VV only) is fetched from Microsoft Planetary Computer via the STAC API plus cloud-optimised GeoTIFF range-reads. Per cell we sample 25 representative pads (deduplicated, stratified across active / recently-completed / older cohorts). We define a ~500m × 500m AOI around each pad, perform **one STAC search per firm-quarter** at the union bounding box, and pull every Sentinel-1 acquisition the search returns. Each scene's COG is opened once and clipped to all 25 pad windows from the in-memory raster.

The classification rule, with `cur_db` the target-quarter mean VV and `base_db` the trailing-four-quarter mean for the same pad:

- **`newly_active`** if `cur_db ≥ base_db + 1.5 dB` AND no FracFocus completion record exists before T;
- **`continuously_active`** if `cur_db ≥ base_db + 0.5 dB` AND a completion exists within prior eight quarters;
- **`idle`** otherwise.

Thresholds (1.5 dB / 0.5 dB) are anchored to Ben-David et al. (2021) and Glaeser, Olsen & Welch (2020) and not tuned on 2019–2024. The activity normalisation `relative_activity_delta = absolute_active – trailing_4Q_avg` mitigates size confounding.

7.3 Point-in-time discipline

FracFocus records are masked to `JobStartDate ≤ T`. Sentinel-1 scenes filtered to `datetime ≤ T`. IBES consensus from `tr_ibes`: estimates with `ANNDATS ≤ T` AND (`REVDATS == ANNDATS` OR `REVDATS > T`); median across the active panel is the consensus (REVDATS filter eliminates the survivorship leak in Anderson & Akbas 2020). GDELT articles filtered to publish dates ≤ T-14. EIA WTI uses prints publicly available before T. Earnings dates use IBES `ANNDATS_ACT` as a documented proxy.

7.4 Consensus-anchored revenue forecast

Agent 2's deterministic core computes the forecast as a satellite-adjusted overlay on the IBES prior:

```
drilling_signal = relative_activity_delta / max(trailing_4Q_avg, 1) forecast_revenue
= consensus_revenue × (1 + α × clip(drilling_signal, –1, +1))
```

α is frozen ex ante: **α = 0.10** for the headline, **α = 0.0** as the no-satellite ablation. This reframes the empirical question from "did our model beat consensus" — a claim we do not make — to "did the satellite signal justify a bounded, principled disagreement with consensus." When IBES coverage is unavailable, a documented fallback model is used and flagged in the manifest.

7.5 Trade-decision logic

A long entry requires Agent 3's `divergence_class` to be `modest_beat` (≥ +5%) or `strong_beat` (≥ +15%). Under the $\alpha = 0.10$ cap the maximum achievable divergence is +10%, so `modest_beat` is the operative entry signal. Agent 4's `gdelt_disclosed` finding downgrades conviction one tier. The lookup maps Arbiter tier to size (15% / 10% / 5% / 0%). The Agent-3 gate short-circuits to `no_trade` whenever divergence is not in `{modest_beat, strong_beat}`; when it fires, Agents 4 and 5 do not run.

Trades enter at the close on T = earnings_date − 14 calendar days and exit on the second trading day strictly after earnings. Returns are total-return-with-dividends; round-trip cost 30 bps; long-only with 15% per-cell cap.

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8. Portfolio Construction and Risk Management

Portfolio rules are deliberately rigid to prevent the multi-agent system from drifting into ad-hoc allocation. Sizing is a deterministic conviction-to-size lookup, so the Investment Board's only discretion is *which* trades to take, not *how much* capital to allocate.

Sizing. The Arbiter assigns each long candidate one of four conviction tiers — `high`, `medium`, `low`, `none`. The lookup is fixed at `high → 15%`, `medium → 10%`, `low → 5%`, `none → 0%`. The 15% per-name cap binds even when conviction would otherwise warrant more, preventing single-name concentration regardless of LLM reasoning quality.

Concurrency cap (documented, non-binding in this sample). The runner manifest records `max_names: 8` as a documented portfolio-construction parameter. In the empirical 2019–2024 sample the cap never binds — the maximum simultaneous longs in any single quarter is 2021-Q3 with 8 trades (every then-eligible firm), exactly at the cap. The inference pipeline does not enforce the cap; the headline P&L in §9 counts every long the system generated. We disclose this gap in §12.6 rather than enforce the cap retroactively.

Cash sleeve. Unallocated capital is held in BIL (SPDR Bloomberg 1–3 Month T-Bill ETF) rather than zero-yield cash. BIL is the system's only non-WRDS price input; it keeps the cash sleeve tradable, free, and total-return-comparable to equity legs.

Trade timing. All entries open at the close on T-14 (two weeks before earnings) and close on the second trading day after the earnings report (T+2). The holding window is short enough that we treat WTI level as approximately constant and long enough to capture the earnings-surprise reaction.

Costs. A flat 30 bps round-trip cost is applied to entry value. We do not separately model bid-ask spread, market impact, borrow, or earnings-event slippage; the long-only restriction makes these simplifications defensible.

Risk attribution. Because sizing is deterministic given conviction tier, realised return decomposes cleanly across the agent stack: name selection is the LLM-driven step (Investment Board plus upstream agents), magnitude is the deterministic step. The attribution analysis in §11 leverages this separation.

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9. Results: Headline

9.1 Aggregate metrics, 2019–2024

Across **Q1 2019 – Q4 2024 (six full calendar years, 200 firm-quarter cells)**, the system produced 51 long entries. All cells use an identical rebalance schedule (T = earnings_date − 14 days, exit second trading day after earnings), 30 bps round-trip cost, and \$1M starting capital. The full per-trade ledger is in Appendix A.

Table 9.1 — Aggregate metrics, 2019–2024:

Metric	Value
Cells evaluated	200
Long trades fired	**51**
Wins / Losses	**27W / 24L**
Hit rate	**52.9%**
Mean net return per trade	**+0.18%**
Median net return per trade	+0.88%
Total net P&L on \$1M	**+\$6,634** (+0.66%)
Best trade	SM 2021-Q3 (+24.45% / +\$24,453)
Worst trade	MTDR 2020-Q3 (–21.51% / –\$21,509)
Per-trade Sharpe (quarterly)	0.021
Max drawdown	–47.06%

Per-year:

Year	Cells	Trades	W/L	Hit	P&L on \$1M
2019	24	2	2/0	100.0%	+\$5,846
2020	28	3	2/1	66.7%	-\$464
2021	30	16	10/6	62.5%	+\$4,794
2022	38	9	5/4	55.6%	+\$10,210
2023	40	9	4/5	44.4%	+\$12,562
2024	40	12	4/8	33.3%	-\$26,313

Total	**200**	**51**	**27/24**	**52.9%**	**+\$6,634**

9.2 Distribution of returns

The ledger has a diffuse return distribution rather than one dominated by outliers. Largest contributors: SM 2021-Q3 (+\$24,453), OVV 2023-Q2 (+\$15,504), SM 2022-Q3 (+\$12,605), EOG 2022-Q3 (+\$10,605). Largest losses: MTDR 2020-Q3 (–\$21,509), MTDR/OVV 2021-Q2 (–\$13,400 each), EOG 2022-Q4 (–\$10,059). No single trade exceeds 2.5% of capital. Excluding the single best trade leaves 50 trades at -\$17,819; excluding the two best leaves 49 at -\$33,323 — the aggregate is moderately resilient to single-trade exclusion but is not robust beyond that, characteristic of a small-sample result.

The 2020-Q1 cohort contributes a single trade: OVV 2020-Q1 (+16.90% / +\$16,900). SM 2020-Q1 does not fire because its 2019 trailing baseline of 9.5 active pads places the satellite-anchored forecast 0.5% *below* consensus, classified `in_line`.

9.3 Primary test (H1)

Pre-registered: one-sided 5%-level firm-clustered bootstrap of hit rate against H0 = 50%. Observed hit rate **0.529**; bootstrap **p = 0.328** (exact-binomial cross-check: 0.390); 95% CI **[39.5%, 65.1%]**. The test **fails to reject H0 = 50%**. The firm-clustered 95% CI on mean net return [-1.83%, +1.53%] straddles zero — the data are also consistent with no average edge. The quarter-block bootstrap yields [33.3%, 67.8%], wider as expected. The directional estimate sits modestly above coin-flip but the test does not support a sharp claim of beating it.

9.4 Secondary test (H2)

H2 compares Strategy 1 against Strategy 3 (analyst-revision follower). Strategy 3 was run on 2024 only, so the like-for-like comparison restricts Strategy 1 to its 2024 trades (n = 12, **33.3%**, P&L **-\$26,313**). Strategy 3 on 2024 is 27.8% over 18 trades. Pre-registered threshold +3 pp; observed gap +5.5 pp — exceeds the threshold on hit rate. **Caveat**: Strategy 1 exceeds Strategy 3 on hit rate but Strategy 1's 2024 P&L is more negative. H2 is technically supported on the pre-registered metric but does not amount to a positive return claim — both baselines lost money in 2024.

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10. Results: Cross-Strategy Comparison

10.1 Per-strategy hit-rate ranking — 2024

For the **Q1 2024 – Q4 2024** real-data sub-window, all signals are computed from real data (real EIA WTI, real EIA-DPR rig count, real Sentinel-1 SAR, real IBES consensus, real CRSP+Yahoo prices). Strategies 3–10 are well-known baselines whose pre-earnings hit rate is documented in the empirical asset-pricing literature.

Rank	Strategy	Hit rate	n_trades
1	Strategy 7 — XLE buy-and-hold	100%	1
2	**Strategy 1 — full multi-agent**	**33.3%**	**12**
3	Strategy 9 — value composite	33.3%	15
4	Strategy 6 — equal-weight universe	32.5%	40
5	Strategy 4 — WTI 3-month momentum	30.0%	20
6	Strategy 5 — Permian rig count	30.0%	10
7	Strategy 8 — 12-1 momentum	28.6%	14
8	Strategy 3 — analyst-revision follower	27.8%	18
9	Strategy 10 — quality composite	18.8%	16
—	Strategy 2 — no-news ablation	not run	—

All classical single-signal baselines earned 2024 hit rates between 19% and 33% — well below coin flip, reflecting 2024's particular regime (rangebound WTI, elevated equity-sector volatility, anti-correlation between pre-earnings drift and several conventional signals). Strategy 7 (XLE) is a single buy-and-hold position. Strategy 1 fired 12 trades in 2024 with a hit rate of 33.3% — tied with Strategy 9 for second place among non-passive strategies, ahead of Strategy 3 by +5.5 percentage points (the H2 comparison in §9.4) — but Strategy 1's 2024 P&L was -**$26,313**, the system's worst calendar year. Outperforming a poorly-performing baseline on hit rate is not equivalent to making money.

10.2 Risk and drawdown

We do not run a Sharpe-difference test against another strategy; small samples make Sharpe noisy. Strategy 1's 2024 drawdown is driven by a string of eight losing entries; the deeper in-universe baseline drawdowns (Strategy 4: −41%; 5: −10%; 6: −62%; 10: −47%) reflect always-long exposure during a rangebound year. Strategy 1's gating step partially reduces drawdown relative to those baselines, but it still fires enough 2024 entries for several losing cells to contribute to drawdown.

10.3 Scope of claim

The ranking is descriptive — pre-registered tests are in §§9 and 11. In 2024, every single-signal baseline (including Strategy 1) lost money or barely broke even; Strategy 1 lost less per trade than several baselines but more in aggregate because it fired more trades.

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11. Ablations and Robustness

The two pre-registered ablations (§§11.1–11.2) are mechanically guaranteed by the deterministic Agent-3 gate. Diagnostics and sensitivities are reported in summary; full tables are in Appendix B.

11.1 RQ2 — $\alpha = 0$ ablation

Setting α to zero makes Agent 2's forecast equal IBES consensus exactly. Agent 3 classifies every such cell as `in_line`; the gate short-circuits to `no_trade`. **Result: zero long trades** across 2019–2024. With $\alpha = 0$, Agents 4 and 5 never fire, so the marginal contribution of the satellite-anchored $\alpha = 0.10$ weighting is the difference between the headline trade count and zero. RQ2 confirms the system's trade-firing mechanism is, in a strict mechanical sense, attributable to the satellite signal, and rules out the failure mode where Agents 4–5 generate spurious "long" signals from qualitative inputs alone.

11.2 RQ3 — no-satellite ablation

Forcing Agent 1 to emit all-idle classifications drives `drilling_signal` to zero, so the forecast collapses to consensus and Agent 3 returns `in_line`. **Result: zero long trades.** Any non-trivial hit rate at the headline must therefore be a statement about the satellite signal *in conjunction with* the LLM scaffolding, not the scaffolding alone.

11.3 Agent-3 deterministic gate

Agent 3 computes class, percentage, and confidence deterministically in Python after the LLM call (which produces only explanatory text); a schema validator rejects any output where the LLM's class disagrees with the rule. This is the architectural mechanism that makes §§11.1–11.2 mechanically guaranteed.

11.4 Defensive fallbacks and provider routing

Agent 4 has a defensive fallback (permissive default on JSON/runtime errors). The headline routes every LLM call through OpenAI (`gpt-4o-mini` for Agents 2/3/4 and Bull/Bear; `gpt-5-mini` for the Arbiter); total cost ~\$1.00–\$2.00 with caching. The deterministic core is provider-independent. Threshold sensitivity over the SAR change-detection rule is deferred.

11.5 Revenue-mechanism diagnostic

Joining each cell's Agent-3 class with Compustat `saleq` actual revenue: **41 of 51 long trades (80.4%) corresponded to actual revenue beats**; 22 of those 41 became stock-return wins (53.7%). The 10 cells where the satellite-anchored forecast was directionally wrong on revenue produced 5W / 5L. A correctly-forecast revenue beat is necessary but not sufficient for a positive 2-trading-day exit, since the stock reaction integrates EPS, capex guidance, hedge-book commentary, and the prevailing oil-price regime. The diagnostic does not change H1.

11.6 WTI stress-veto sensitivity

A pre-registered point-in-time veto: for each long trade with entry date T, **block the entry if `wti_4w_return(T) ≤ –10.0%`**. Sizing, costs, and exit are unchanged.

The veto blocks one trade — OVV 2020-Q1 (4w WTI = −16.8%, baseline +\$16,900). Post-veto: 50 trades, 26W / 24L (52.0%), total **-\$10,266** (vs +\$6,634 baseline); bootstrap p = 0.384; 95% CI on hit rate [38.1%, 64.8%].

Interpretation. The pre-registered veto is **not protective**. The −10% threshold is misaligned with the system's signal direction in the 2020-Q1 dislocation: OVV 2020-Q1 opened *after* the WTI negative-pricing event of 2020-04-20 and exited in early May as oil partially recovered, capturing a regime-conditional mean-reversion the veto would have foreclosed. We do not retune. The qualitative finding is that a single-threshold WTI guardrail removes a gain, not a loss.

11.7 Summary

(i) Remove the satellite (or set $\alpha = 0$) $\rightarrow$ system mechanically stops trading. (ii) The Agent-3 gate eliminates LLM boundary noise. (iii) The deterministic core is provider-independent. (iv) Revenue mechanism directionally correct on 80% of traded cells; trade-monetisation adds variance independent of revenue accuracy. (v) Pre-registered WTI veto **removes a gain, not a loss**, shifting aggregate from +\$6,634 to -\$10,266. The constellation argues for caution about the +\$6,634 headline; mean return per trade (+0.18%) is within transaction-cost noise.

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12. Limitations and Threats to Validity

12.1 SAR change-detection rule

The Sentinel-1 RTC ingestion is real and the per-pad backscatter time series are real radar pixels — but the change-detection rule is intentionally simple: a fixed +1.5 dB / +0.5 dB threshold on quarterly mean VV backscatter, anchored to Ben-David et al. (2021) and Glaeser, Olsen & Welch (2020). The rule has no CV/DL component, no polarimetric decomposition, and no coherent change detection on SLC products. A more sophisticated pipeline — a U-Net or fine-tuned earth-observation foundation model such as NASA-IBM Prithvi — would catch finer signals and emit per-pad probabilities. We treat this as the system's most consequential simplification.

12.2 Sample size, IBES proxy, LLM determinism

The full backtest produces 51 long trades across 200 cells (the gate suppresses ~75%). With n = 51 the firm-clustered bootstrap fails to reject H0 = 50% (p = 0.328); the firm-clustered 95% CI on mean return [-1.83%, +1.53%] straddles zero.

The IBES consensus uses `ANNDATS_ACT` as a proxy for pre-T-14 announcement-date provability (WRDS Wall Street Horizon not subscribed). U.S. E&P firms pre-announce calendars 30+ days ahead, so the proxy is unbiased in expectation but subtly biased when announcement dates shift. The survivorship leak (Anderson & Akbas 2020) is mitigated by the `(REVDATS == ANNDATS) OR (REVDATS > T)` filter.

LLM determinism: all calls run at temperature 0.0 with a per-call cache. OpenAI does not guarantee bit-for-bit determinism, but variance is small for constrained-JSON outputs and the deterministic-numerical-core means trades are insensitive to small qualitative-text changes.

12.3 Universe, architecture, scope

The ten-firm universe is selected on Permian concentration, correlated with mid-/small-cap status; the universe was selected knowing all ten firms exist as of 2025, itself a forward-looking selection. CRGY produces zero FracFocus Permian disclosures. The 25-pad sample is bandwidth-limited at ~12 hours wall-clock. The deterministic conviction-to-size lookup preserves traceability at the cost of bounded achievable Sharpe. Strategies 2–10 are 2024-only; extending them is the most direct future-work item.

12.4 Recovery-regime over-firing and signal saturation

The backtest's most consequential structural limitation is **2021 over-firing**: 16 long trades in 2021 (post-COVID recovery) vs 2–12 in every other year. **13 of 16 (81%) reach `modest_beat` via a clipped `drilling_signal = +1.0`**, pinning forecast divergence at exactly +10.0%. Mechanism: trailing-4Q active-pad averages in 2021 are dominated by the COVID 2020 trough (most pads idle by Agent 1's rule), so baselines are low (1.25–7.75). 2021 active counts (7–23) divided by those baselines produce raw signals of 1.4–7.5, all clipped at +1.0.

The clip prevents runaway forecasts when baselines approach zero, but the consequence is that the gate cannot distinguish a modest recovery from an explosive one when the baseline collapsed. 13 of 16 cells share an identical +10.0% divergence number, so Agent-3 loses rank-ordering within the cohort. This explains why the 2021 cohort produces 16 trades at 62.5% hit but only +0.50% mean — the gate fires broadly without ranking. We do not modify the clip or baseline formula post-hoc; §13 lists "regime-aware sizing" as future work. Excluding the 2021 cohort leaves

35 trades at 17W / 18L (48.6%) and aggregate +\$1,840.

12.5 External validity

Regime. The 2021 recovery cohort is the largest single-year trade-count contributor (16 of 51) but not aggregate-P&L. Generalising to future regime breaks is not supported by the historical backtest — the over-firing pattern is specific to the post-COVID setup and the design's failure mode in that setup. **Crowding.** If the satellite signal becomes commercially available at scale, the edge will compress within 12–36 months per the signal-decay literature.

12.6 Documented portfolio rules versus enforced behaviour

The runner manifest records `max_names: 8` as a documented portfolio-construction parameter. In this sample the cap **never binds** — the maximum simultaneous longs in any quarter is 8 (2021-Q3, all eight then-eligible firms). The cap is not enforced in the inference layer; the headline P&L counts every long the system generated. Because the cap never binds, the headline is unaffected. §8 describes the cap as a documented parameter that did not bind in this sample rather than a binding rule with priority semantics.

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13. Conclusion

We documented a multi-agent trading system for Permian E&P in which satellite-derived drilling activity feeds a deterministic revenue forecast assessed by an LLM-based investment board. Deterministic financial logic runs in code; qualitative outlook synthesis, news verification, and Bull/Bear/Arbiter debate run in language models.

This decomposition addresses the principal failure mode of agentic-AI trading: delegation of unverifiable numerical decisions to language models, evaluated by ex-post P&L rather than ex-ante calibration. Our system avoids that failure mode by construction — every long trade is traceable to a small set of explicit numerical inputs, every LLM agent's marginal contribution measurable by ablation. The $\alpha = 0$ and no-satellite ablations both produce zero long trades.

Empirical results on the real-data 2019–2024 window: **51 long trades**, 27W / 24L (52.9%), +0.18% mean net return per trade, median +0.88%, total **+\$6,634 on \$1M starting capital** (+0.66% of capital). With $n = 51$ the firm-clustered bootstrap fails to reject $H0 = 50\%$ ($p = 0.328$, 95% CI on hit rate [39.5%, 65.1%]); the firm-clustered 95% CI on mean return [-1.83%, +1.53%] straddles zero. No single trade dominates the aggregate; the largest single contributor is SM 2021-Q3 (+\$24,453, ~2.4% of capital). The pre-registered WTI stress veto blocks one trade (OVV 2020-Q1, +\$16,900) and shifts the aggregate to -\$10,266, showing that a single-threshold macro guardrail removes a gain rather than a loss in this window.

The most consequential structural finding is **2021 over-firing** (§12.4): the system fires 16 long trades in 2021, 13 of which clip at `drilling_signal = +1.0` and reach an identical +10.0% divergence number. The clip works as designed but collapses signal differentiation across the recovery cohort. We disclose rather than tune.

We are explicit about what the paper does not claim: profitability after realistic execution costs (the +0.18% mean return is well within the 30-bps round-trip band), statistical sufficiency at $n = 51$, optimality of the SAR rule, or that the IBES `ANNDATS_ACT` proxy is fully look-ahead-clean. The contribution is methodological — a multi-agent design pattern with deterministic numerical core, end-to-end real-data ingestion, and ablation-validated component contribution — portable to other alt-data-meets-fundamentals settings.

Four future-work avenues. (1) Upgrade SAR change detection to a CV/DL pipeline. (2) Regime-aware sizing or gating that conditions on trailing-baseline magnitude rather than the clipped signal, restoring rank-ordering in recovery cohorts. (3) Extend Strategies 2–10 to the full 2019–2024 window. (4) Implement the documented 8-name concurrency cap or drop it from the documentation.

The result is statistically modest: the system produces a directionally-positive signal whose marginal contribution is mechanically measurable, but the aggregate is small and not significant against a coin-flip null. The contribution is a discipline of separating deterministic from LLM roles, pre-registering tests, and running ablations whose results could falsify the central thesis. These disciplines are most important when an LLM is in the pipeline and most important when the result is unflattering.

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Appendix

A. Per-trade ledger, Strategy 1, 2019–2024

Each row is a single long entry. Entry price = closing price on T-14, exit price = closing price on the second trading day after the earnings announcement. Net return is after the 30 bps round-trip transaction cost. The full machine-readable ledger is at `runs/inference/strategy01_trades.csv`.

#	Cell	Size	Tier	Entry T-14	Earnings	Exit (~T+2)	Net return	W/L
1	SM 2019-Q2	0.05	low	\$9.35 (2019-07-18)	2019-08-01	\$9.59	+2.27%	W
2	DVN 2019-Q3	0.05	low	\$20.67 (2019-10-22)	2019-11-05	\$22.68	+9.42%	W
3	OVV 2020-Q1	0.10	medium	\$5.00 (2020-04-23)	2020-05-07	\$5.86	+16.90%	W
4	MTDR 2020-Q2	0.10	medium	\$8.55 (2020-07-14)	2020-07-28	\$8.93	+4.14%	W
5	MTDR 2020-Q3	0.10	medium	\$8.77 (2020-10-13)	2020-10-27	\$6.91	-21.51%	L
6	OVV 2021-Q1	0.10	medium	\$24.21 (2021-04-15)	2021-04-29	\$24.94	+2.72%	W
7	SM 2021-Q1	0.10	medium	\$17.42 (2021-04-15)	2021-04-29	\$16.52	-5.47%	L
8	FANG 2021-Q1	0.10	medium	\$77.67 (2021-04-19)	2021-05-03	\$83.12	+6.72%	W
9	OVV 2021-Q2	0.10	medium	\$30.11 (2021-07-13)	2021-07-27	\$26.07	-13.72%	L
10	MTDR 2021-Q2	0.10	medium	\$35.54 (2021-07-13)	2021-07-27	\$30.90	-13.36%	L
11	SM 2021-Q2	0.10	medium	\$20.42 (2021-07-15)	2021-07-29	\$18.72	-8.63%	L
12	FANG 2021-Q2	0.10	medium	\$73.19 (2021-07-19)	2021-08-02	\$76.42	+4.11%	W
13	OXY 2021-Q2	0.10	medium	\$25.37 (2021-07-20)	2021-08-03	\$26.19	+2.93%	W

14	MTDR 2021-Q3	0.10	medium	\$42.39 (2021-10-12)	2021-10-26	\$42.32	-0.47%	L
15	**SM 2021-Q3**	0.10	medium	\$29.37 (2021-10-14)	2021-10-28	\$36.64	**+24.45%**	W
16	FANG 2021-Q3	0.10	medium	\$109.26 (2021-10-18)	2021-11-01	\$111.81	+2.03%	W
17	DVN 2021-Q3	0.10	medium	\$40.30 (2021-10-19)	2021-11-02	\$42.36	+4.81%	W
18	OVV 2021-Q3	0.10	medium	\$39.27 (2021-10-19)	2021-11-02	\$35.74	-9.29%	L
19	CTRA 2021-Q3	0.10	medium	\$21.41 (2021-10-20)	2021-11-03	\$21.58	+0.49%	W
20	EOG 2021-Q3	0.05	low	\$90.90 (2021-10-21)	2021-11-04	\$96.99	+6.40%	W
21	OXY 2021-Q3	0.10	medium	\$32.80 (2021-10-21)	2021-11-04	\$34.29	+4.24%	W
22	DVN 2022-Q3	0.10	medium	\$69.77 (2022-10-18)	2022-11-01	\$70.73	+1.08%	W
23	EOG 2022-Q3	0.10	medium	\$131.59 (2022-10-20)	2022-11-03	\$145.94	+10.61%	W
24	SM 2022-Q3	0.10	medium	\$42.85 (2022-10-20)	2022-11-03	\$48.38	+12.61%	W
25	FANG 2022-Q3	0.05	low	\$153.26 (2022-10-24)	2022-11-07	\$157.55	+2.50%	W
26	OXY 2022-Q3	0.10	medium	\$71.09 (2022-10-25)	2022-11-08	\$70.48	-1.16%	L
27	OVV 2022-Q3	0.10	medium	\$50.94 (2022-10-25)	2022-11-08	\$53.85	+5.41%	W
28	DVN 2022-Q4	0.05	low	\$63.24 (2023-01-31)	2023-02-14	\$55.72	-12.19%	L
29	EOG 2022-Q4	0.10	medium	\$127.16 (2023-02-09)	2023-02-23	\$114.75	-10.06%	L
30	OVV 2022-Q4	0.05	low	\$48.38 (2023-02-13)	2023-02-27	\$45.21	-6.85%	L

31	SM 2023-Q1	0.10	medium	\$30.96 (2023-04-13)	2023-04-27	\$28.04	-9.73%	L
32	CTRA 2023-Q1	0.10	medium	\$25.56 (2023-04-20)	2023-05-04	\$24.86	-3.04%	L
33	MTDR 2023-Q2	0.10	medium	\$53.84 (2023-07-11)	2023-07-25	\$53.24	-1.41%	L
34	OVV 2023-Q2	0.10	medium	\$39.80 (2023-07-13)	2023-07-27	\$46.09	+15.50%	W
35	FANG 2023-Q2	0.10	medium	\$135.47 (2023-07-17)	2023-07-31	\$146.79	+8.06%	W
36	SM 2023-Q2	0.10	medium	\$34.76 (2023-07-19)	2023-08-02	\$38.33	+9.97%	W
37	FANG 2023-Q3	0.10	medium	\$165.32 (2023-10-23)	2023-11-06	\$155.97	-5.96%	L
38	CTRA 2023-Q3	0.10	medium	\$28.40 (2023-10-23)	2023-11-06	\$26.88	-5.65%	L
39	OXY 2023-Q4	0.10	medium	\$57.57 (2024-01-31)	2024-02-14	\$60.52	+4.82%	W
40	DVN 2024-Q1	0.10	medium	\$51.67 (2024-04-17)	2024-05-01	\$50.54	-2.49%	L
41	OVV 2024-Q2	0.10	medium	\$48.46 (2024-07-16)	2024-07-30	\$44.48	-8.51%	L
42	FANG 2024-Q2	0.10	medium	\$204.68 (2024-07-22)	2024-08-05	\$191.38	-6.80%	L
43	PR 2024-Q2	0.10	medium	\$15.67 (2024-07-23)	2024-08-06	\$14.53	-7.58%	L
44	SM 2024-Q2	0.05	low	\$44.50 (2024-07-24)	2024-08-07	\$43.11	-3.42%	L
45	MTDR 2024-Q3	0.10	medium	\$53.17 (2024-10-08)	2024-10-22	\$52.40	-1.75%	L
46	SM 2024-Q3	0.10	medium	\$43.52 (2024-10-17)	2024-10-31	\$40.94	-6.23%	L
47	CTRA 2024-Q3	0.10	medium	\$24.07 (2024-10-17)	2024-10-31	\$23.07	-4.45%	L

48	PR 2024-Q3	0.10	medium	\$13.85 (2024-10-23)	2024-11-06	\$14.92	+7.43%	W
49	OVV 2024-Q3	0.05	low	\$39.81 (2024-10-24)	2024-11-07	\$42.69	+6.93%	W
50	OXY 2024-Q3	0.10	medium	\$50.08 (2024-10-29)	2024-11-12	\$50.67	+0.88%	W
51	OVV 2024-Q4	0.10	medium	\$41.24 (2025-02-12)	2025-02-26	\$41.95	+1.43%	W

B. Signal-confidence score specification

A 0–100 deterministic score composes five 0–20 sub-scores from inputs already exposed by upstream agents:

1. **SAR activity strength** — `share_active`, binned 0/5/10/15/20 by 0.0/0.15/0.30/0.45/0.60 thresholds.
2. **SAR signal newness** — `n_newly_active / max(n_active, 1)`, binned 0/5/10/15/20.
3. **QoQ activity delta** — `relative_activity_delta`, binned by sign and magnitude (above-baseline gets higher score).
4. **Consensus tightness** — inverse coefficient of dispersion of the active-analyst panel; tighter = higher score.
5. **Analyst panel breadth** — `n_analysts_at_T_minus_14`, binned 0/5/10/15/20 by 3/6/10/15+ analysts.

Tier mapping: `score ≥ 70 → high`, `40 ≤ score < 70 → medium`, `< 40 → low`. WTI/oil-regime is intentionally excluded so that the score and the WTI veto (§11.6) do not double-count.

The score is logged in `runs/inference/signal_confidence.csv` for the 200 cells of the backtest. It does not enter H1 or any pre-registered test; it is preserved in the appendix as a per-cell quality annotation. The dominant signal-quality story is the 2021 over-firing pattern documented in §12.4 (13 of 16 2021 longs share an identical clipped +1.0 drilling signal and a +10.0% divergence number), which the score's `relative_activity_delta` and `consensus tightness` sub-scores cannot rank-order.

C. Software, data, and computational resources

- **IBES Detail-History** (`tr_ibes`, sales/revenue measure code SAL): WRDS subscription, accessed Q1 2026.
- **Compustat quarterly fundamentals** (`comp.fundq`): WRDS subscription.
- **CRSP daily stock files**: WRDS subscription, through 2024-12-31.
- **yfinance Adj Close**: open-source, used to extend CRSP through 2025.
- **EIA weekly WTI spot**: U.S. Energy Information Administration public release.
- **EIA Drilling Productivity Report Permian rig count**: EIA public release.
- **FracFocus public bulk CSV**: https://fracfocusdata.org/digitaldownload/FracFocusCSV.zip.
- **Microsoft Planetary Computer Sentinel-1 RTC**: https://planetarycomputer.microsoft.com (free, no auth).
- **GDELT 2.0 DOC API**: https://www.gdeltproject.org/.

- **Texas Railroad Commission and New Mexico Oil Conservation Division** permit data: public records via state portals.
- **LLM provider**: OpenAI API. `gpt-4o-mini` for Agent 2 outlook, Agent 3 reasoning text, Agent 4 verification, Agent 5 Bull/Bear; `gpt-5-mini` for the Agent 5 Arbiter.

D. Reproducibility

Every backtest run writes a `manifest.json` recording: SHA256 of all input CSVs (FracFocus permit dump, EIA WTI weekly, EIA-DPR Permian rig count, IBES Detail-History, GDELT cache index, Sentinel-1 firm-quarter aggregate cache index); the SAR-mode flag (`real_sentinel1` for the headline run); the change-detection thresholds (1.5 dB activation, 0.5 dB sustained); the trailing-baseline coefficient (0.3); the consensus-anchor `α`; the per-agent provider and model identifier (and version where the provider exposes it); prompt-file SHAs; the Python version; and the platform identifier.

The LLM-call cache is keyed on `(prompt_sha, input_sha, model_id, model_version, temperature)` and is included in the supplementary materials; combined with the per-pad Sentinel-1 backscatter cache (`phase1/output/sentinel1_cache/`), re-execution is essentially free up to provider determinism limits.

To reproduce the headline run:

```
export FIN580_SAR_MODE=real_sentinel1 export FIN580_SAR_PADS_PER_OP=25 # Execute
Strategy 1 over the six yearly windows from 2019Q1-2019Q4 through 2024Q1-2024Q4, #
then regenerate the inference rollups from those yearly run directories.
```

The per-trade ledger, manifest, and caches are sufficient for an external reviewer to spot-check any trade. Caches and manifests are available on request.